Abhyaung Deepak Havale

AI Engineer

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SUMMARY

AI/ML Engineer with around 4 years of experience designing, developing, and deploying production-grade machine learning and generative AI solutions across healthcare, finance, and enterprise domains. Expertise in end-to-end ML pipelines, predictive modeling, NLP, computer vision, and large language models (LLMs), leveraging TensorFlow, PyTorch, Hugging Face Transformers, LangChain, LlamaIndex, and OpenAI APIs. Strong proficiency in MLOps practices using AWS SageMaker, Azure ML, Docker, Kubernetes, Apache Airflow, MLflow, PySpark, and SQL (PostgreSQL, Snowflake, BigQuery) to deliver scalable, governed, and cost-efficient models that drive measurable business impact.

SKILLS

Programming & Scripting: Python, SQL, Pandas, NumPy, PySpark, OpenCV

Machine Learning & AI: TensorFlow, PyTorch, Scikit-learn, CNN, Transfer Learning, Logistic Regression, Random Forest, XGBoost, LightGBM, Gradient Boosting, BERT, Hugging Face Transformers, SpaCy, OpenAI GPT, Claude, LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Onnx, DeepSpeed, ElevenLabs

Natural Language Processing: NER, Text Classification, Clinical Entity Extraction, OCR, Document Classification, Medical Terminology Processing, LLM Prompt Engineering & Fine-Tuning

Data Engineering & MLOps: Apache Airflow, AWS Glue, AWS S3, AWS SageMaker, MLflow, ETL Pipelines, Data

Preprocessing, Feature Engineering, Model Retraining Automation, Pulumi, Kubernetes, Docker, Serverless (AWS Lambda), GitHub Actions (CI/CD)

Model Deployment & APIs: Flask, FastAPI, RESTful API Services, ERP System Integration

Data Visualization & Reporting: Power BI, Tableau, AWS QuickSight, Matplotlib, Seaborn, Risk Scorecards, Model Monitoring Dashboards, KPI Tracking

Compliance & Governance: SOX, PCI DSS, FFIEC, FDA AI Guidelines, GDPR, Data Privacy & Security, Bias Mitigation, Transparent AI Practices

EXPERIENCE

AI Engineer, Freddie Mac, USA **Jan 2025 – Present**

- Led the design and tuning of an LLM-based candidate ranking system, analyzing profiles and cutting response time by 83%.
- Designed and improved PostgreSQL and Elasticsearch setups to support fast searches with complex filters and synonyms.
- Set up Docker containers for microservices and used GCP Cloud Run with blue/green deployments, reducing deployment time to 8 minutes and maintaining 99.899% uptime.
- Created CI/CD pipelines in GitHub Actions for testing, staging, and production with automatic error handling and rollback.
- Developed RAG (Retrieval-Augmented Generation) system using Hugging Face Transformers (sentence-transformers + cross-encoder) and FAISS vector store to provide contextual answers from internal policy and contract repositories, improving query accuracy by 38%.
- Used GCP Cloud Functions to import and clean data from Greenhouse/Lever, lowering cloud costs while keeping data fresh.
- Managed Airflow DAGs that triggered PySpark tasks on GCP Dataproc, making data workflows 35% more reliable.
- Built two major features in two sprints: backend logic for candidate scores & advanced APIs for background jobs and polling.

Machine Learning Engineer, Hexaware Technologies, INDIA **Dec 2020 – Aug 2023**

- Designed and deployed machine learning models to analyze customer interactions across digital channels, improving customer segmentation accuracy by 34%.
- Developed and fine-tuned NLP models using spaCy and BERT to perform sentiment analysis and topic modeling on support ticket data and built automated data pipelines using Python, Pandas, and Apache Airflow to ingest and preprocess unstructured data from various CRM systems.
- Implemented predictive models using scikit-learn and XGBoost to forecast customer churn, increasing retention strategy effectiveness by 21% and containerized ML workflows with Docker and deployed RESTful inference services using Flask and AWS Lambda for scalable integrations.
- Collaborated with cross-functional teams to define model evaluation metrics, achieving 90%+ F1-score across production use cases and used MLflow for experiment tracking and Git for version control, while managing model deployment cycles via AWS SageMaker.
- Ran Agile meetings like sprint planning and daily standups using Jira and Confluence; used GitHub PRs and Linear to stay on track with deliverables.

EDUCATION

Master's in Computer Science, Pace University, New York, USA **Sep 2023 – May 2025**

BSc Computer Science, Modern College of Arts, Commerce & Science, Pune, INDIA **Jul 2019 – Dec 2021**

CERTIFICATION

AWS Cloud Practitioner